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ASSIGNMENT 8

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HPS, Chapter 2: Twelve Exercises

From the textbook: 8, 9, 10, 11, 14, 16, 29, 38, 54, 55, 59(a), 61.

The exercises are HPS's, from the end of Chapter 2 (pp. 46–51), reproduced verbatim. Notation follows the book where it has one: $B(\theta) = 1/\{nE[(\partial \log f(X \mid \theta)/\partial \theta)^2]\}$ is the lower bound of its Formula (1) (p. 17), $e(\delta) = B(\theta)/V(\delta)$ the efficiency of Definition 2 (p. 17), $\mathcal{L}$ the loss and $\mathcal{R}$ the risk. Throughout, $X_1, \dots, X_n$ is the sample, $s = \sum_i x_i$, and $\bar{X} = \sum_i X_i/n$.

Exercise 8

Show that $d = \sum_{i=1}^n X_i/n$ is an efficient unbiased estimator of the parameter θ for the Poisson density $f(x \mid \theta) = e^{-\theta}\theta^x/x!$, $x = 0, 1, \dots$, where $X_1, \dots, X_n$ represents a random sample of X .

SOLUTION:

Unbiasedness first. Each X_i has mean θ , so by linearity $E^\theta\{d\} = \frac{1}{n} \sum_i E^\theta\{X_i\} = \theta$ for every θ .

For efficiency, compute the quantity in the bound. From $\log f(x \mid \theta) = -\theta + x \log \theta -$

$\log x!$,

$$\frac{\partial \log f(x \mid \theta)}{\partial \theta} = -1 + \frac{x}{\theta} = \frac{x - \theta}{\theta},$$

and since $E^\theta\{(X - \theta)^2\} = \text{Var}^\theta(X) = \theta$ for the Poisson,

$$E \left[\left(\frac{\partial \log f(X \mid \theta)}{\partial \theta} \right)^2 \right] = \frac{\text{Var}^\theta(X)}{\theta^2} = \frac{1}{\theta}, \quad \text{so} \quad B(\theta) = \frac{1}{n \cdot (1/\theta)} = \frac{\theta}{n}.$$

The estimator's variance is $\text{Var}^\theta(d) = \text{Var}^\theta(X)/n = \theta/n$, because the X_i are independent and variances scale by $1/n^2$ across the sum of n of them. Hence

$$e(d) = \frac{B(\theta)}{\text{Var}^\theta(d)} = \frac{\theta/n}{\theta/n} = 1$$

and d is an efficient unbiased estimator in the book's sense of Definition 2 (p. 17): it attains the lower bound at every θ . This closes a small gap in the text itself, whose worked examples in §2.2 are the normal and the Bernoulli only; the Poisson computation ran through Assignments 5 and 6 and lands here as the missing third example. ■

Exercise 9

(a) What is the best set of a 's to choose in Exercise 4 if $Z = \sum_{i=1}^n a_i X_i$ is to be an unbiased estimator of EX with minimum variance? (b) What a 's should be chosen if the X_i are independent variables with the same mean but different known variances σ_i^2 ?

SOLUTION:

Exercise 4 supplies the constraint. With $\mu = E\{X\}$, linearity gives $E\{Z\} = \sum_i a_i E\{X_i\} = \mu \sum_i a_i$, so Z is unbiased for every μ exactly when

$$\sum_{i=1}^n a_i = 1.$$

(a) With a common variance σ^2 and independence, $\text{Var}(Z) = \sigma^2 \sum_i a_i^2$, so the problem is to minimize $\sum_i a_i^2$ subject to $\sum_i a_i = 1$. No calculus is needed: expand the sum of squared deviations from $1/n$,

$$\sum_{i=1}^n \left(a_i - \frac{1}{n}\right)^2 = \sum_i a_i^2 - \frac{2}{n} \sum_i a_i + n \cdot \frac{1}{n^2} = \sum_i a_i^2 - \frac{2}{n} + \frac{1}{n} = \sum_i a_i^2 - \frac{1}{n},$$

using the constraint in the middle term. The left side is nonnegative and vanishes only when every $a_i = 1/n$, so $\sum_i a_i^2 \geq 1/n$ with equality exactly there:

$$a_i = \frac{1}{n} \text{ for every } i: \text{ the sample mean, with variance } \sigma^2/n$$

(b) Now $\text{Var}(Z) = \sum_i a_i^2 \sigma_i^2$, to be minimized under the same constraint. The Lagrange condition sets $\partial/\partial a_i [\sum_j a_j^2 \sigma_j^2 - \lambda \sum_j a_j] = 2a_i \sigma_i^2 - \lambda = 0$, so $a_i = \lambda/(2\sigma_i^2)$: the weights

must be proportional to the reciprocal variances. The constraint fixes the constant:

$$\boxed{a_i = \frac{1/\sigma_i^2}{\sum_{j=1}^n 1/\sigma_j^2}} \quad \text{with} \quad \text{Var}(Z) = \left(\sum_j 1/\sigma_j^2 \right)^{-1},$$

the variance following on substituting the weights: $\sum_i a_i^2 \sigma_i^2 = \sum_i \sigma_i^{-2} / (\sum_j \sigma_j^{-2})^2 = 1 / \sum_j \sigma_j^{-2}$. (That the critical point is the minimum follows because the objective is a positive-definite quadratic on the constraint plane.) Precise observations get large weights, noisy ones small; precisions add in the denominator. This is the same arithmetic that set the posterior weights in Assignment 7's normal problem, with data replacing the prior as the second voice. ■

Exercise 10

Show that the estimator $Z = (X_1 + X_2)/2$ is an unbiased estimator of EX , where $X_1, \dots, X_n$ represents a random sample of X . Calculate the efficiency of Z if X possesses the density

$$f(x | \theta) = \frac{\exp [-(1/2)(x - \theta)^2]}{\sqrt{2\pi}}.$$

SOLUTION:

Unbiasedness holds for any parent distribution: $E\{Z\} = \frac{1}{2}(E\{X_1\} + E\{X_2\}) = EX$ by linearity.

For the normal density with unit variance, $\text{Var}(Z) = \frac{1}{4}(\text{Var}(X_1) + \text{Var}(X_2)) = \frac{1}{4} \cdot 2 = \frac{1}{2}$, the two observations being independent. The bound uses

$$\log f(x | \theta) = -\frac{1}{2}(x - \theta)^2 - \frac{1}{2} \log(2\pi), \quad \frac{\partial \log f}{\partial \theta} = x - \theta, \quad E[(X - \theta)^2] = 1,$$

so $B(\theta) = 1/(n \cdot 1) = 1/n$, and Definition 2 gives

$$e(Z) = \frac{B(\theta)}{\text{Var}(Z)} = \frac{1/n}{1/2} = \frac{2}{n}$$

The number reads as an accounting of waste: Z throws away $n-2$ of the n observations, and its efficiency is exactly the fraction of the sample it uses. At $n = 2$ it is the sample mean and the efficiency is 1; for larger n an unbiased estimator with variance $1/n$ exists (the full mean), and Z delivers only $2/n$ of that precision. ■

Exercise 11

Let X be a random variable with finite second moment. Show that $E(X - b)^2$, where b is a constant, is minimized by $b = EX$.

SOLUTION:

Write $\mu = EX$, add and subtract it inside the square, and expand:

$$\begin{aligned} E\{(X - b)^2\} &= E\{[(X - \mu) + (\mu - b)]^2\} && \text{(add and subtract } \mu) \\ &= E\{(X - \mu)^2\} + 2(\mu - b) E\{X - \mu\} + (\mu - b)^2 \\ &&& \text{(expand; } \mu - b \text{ is constant)} \\ &= \text{Var}(X) + (\mu - b)^2. && (E\{X - \mu\} = 0) \end{aligned}$$

The first term does not involve b and the second is nonnegative, so the whole expression is minimized exactly when $(\mu - b)^2 = 0$:

$$b = EX, \text{ with minimum value } \text{Var}(X)$$

The finite second moment is what makes every expectation above exist. Small as it is, this fact has been carrying weight all course: it is the reason the best constant predictor is the mean (notes, Thm. 2.8.1), the inner step of the posterior-mean theorem used throughout Assignment 7, and—applied conditionally—the reason the best predictor given data is the conditional expectation, which Exercise 59 below puts to work. ■

Exercise 14

Calculate the mean and variance of the estimator $Z = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n}$ for the parameter θ in

$$f(x \mid \theta) = \frac{e^{-[(x-\mu)^2/2\theta]}}{\sqrt{2\pi\theta}}.$$

Use the fact from Volume I that $(X - \mu)^2/\theta$ possesses a chi-square distribution with one degree of freedom and hence that nZ/θ possesses a chi-square distribution with n degrees of freedom. This assumes that $X_1, \dots, X_n$ represent a random sample of X .

SOLUTION:

Here θ is the variance of a normal with known mean μ . Following the given fact, write $W = nZ/\theta$, so $W \sim \chi_n^2$ with $E\{W\} = n$ and $\text{Var}(W) = 2n$, and $Z = \theta W/n$. Then

$$E^\theta\{Z\} = \frac{\theta}{n} E\{W\} = \frac{\theta}{n} \cdot n = \boxed{\theta} \quad (\text{linearity; } \chi_n^2 \text{ mean})$$

$$\text{Var}^\theta(Z) = \frac{\theta^2}{n^2} \text{Var}(W) = \frac{\theta^2}{n^2} \cdot 2n = \boxed{\frac{2\theta^2}{n}} \quad (\text{scaling squares; } \chi_n^2 \text{ variance})$$

So Z is unbiased for the variance with sampling variance $2\theta^2/n$. (The book's Exercise 15, not assigned, checks this against the bound and finds efficiency 1; Exercise 29 below revisits this same Z and asks a sharper question.) ■

Exercise 16

By differentiating (5) under the integral sign with respect to θ and using the relationship

$$\frac{\partial L}{\partial \theta} = L \frac{\partial \log L}{\partial \theta},$$

show that

$$E \left(\frac{\partial \log f(X | \theta)}{\partial \theta} \right)^2 = -E \left(\frac{\partial^2 \log f(X | \theta)}{\partial \theta^2} \right).$$

This gives an alternative method for calculating the lower bound in (2) when it is applicable.

SOLUTION:

The book's display (5) (p. 19) is the statement that the derivative of the log likelihood has mean zero, written for a single observation as

$$\int \frac{\partial \log f(x | \theta)}{\partial \theta} f(x | \theta) dx = 0 \quad \text{for every } \theta,$$

which itself came from differentiating $\int f dx = 1$ under the integral sign. Differentiate it once more in θ , the product rule acting on the two θ -dependent factors:

$$\begin{aligned} 0 &= \int \frac{\partial^2 \log f(x | \theta)}{\partial \theta^2} f(x | \theta) dx + \int \frac{\partial \log f(x | \theta)}{\partial \theta} \frac{\partial f(x | \theta)}{\partial \theta} dx \\ &\quad \text{(product rule under the integral)} \\ &= \int \frac{\partial^2 \log f}{\partial \theta^2} f dx + \int \left(\frac{\partial \log f}{\partial \theta} \right)^2 f dx \\ &\quad \text{(the given relation, } \partial f / \partial \theta = f \partial \log f / \partial \theta) \end{aligned}$$

$$= \mathbb{E} \left(\frac{\partial^2 \log f(X | \theta)}{\partial \theta^2} \right) + \mathbb{E} \left(\frac{\partial \log f(X | \theta)}{\partial \theta} \right)^2,$$

(both integrals are expectations against f)

and rearranging gives the display:

$$\mathbb{E} \left(\frac{\partial \log f(X | \theta)}{\partial \theta} \right)^2 = - \mathbb{E} \left(\frac{\partial^2 \log f(X | \theta)}{\partial \theta^2} \right)$$

The same computation runs verbatim with the joint likelihood L in place of f , and with sums in place of integrals for discrete families. This is the identity the course text never states in its own §2.2—its bound (2) is defined through the squared first derivative alone—and Assignment 5 verified it model by model; here it arrives in general, from two differentiations of the normalization. DeGroot and Schervish prove it as Theorem 8.8.1 (pp. 515–516). The practical value is as the exercise says: second derivatives are often easier to average than squared first derivatives, since cross terms die. ■

Exercise 29

Find the value of k that minimizes $E(ks^2 - \sigma^2)^2$, where

$$s^2 = \frac{\sum (X_i - \mu)^2}{n},$$

and the X_i represent a random sample of a normal variable. Use the fact that $\sum (X_i - \mu)^2 / \sigma^2$ is a chi-square variable with n degrees of freedom and that the mean and variance of such a variable are equal to n and $2n$, respectively. What does this result show about the requirement of unbiasedness if expected squared error is the basis for comparing estimators, rather than the variance?

SOLUTION:

Write $W = ns^2/\sigma^2 \sim \chi_n^2$, so $s^2 = \sigma^2 W/n$, $E\{W\} = n$, $\text{Var}(W) = 2n$, and $E\{W^2\} = \text{Var}(W) + (E\{W\})^2 = 2n + n^2$. Then

$$\begin{aligned} E\{(ks^2 - \sigma^2)^2\} &= \sigma^4 E\left\{\left(\frac{kW}{n} - 1\right)^2\right\} && \text{(factor out } \sigma^4) \\ &= \sigma^4 \left[\frac{k^2}{n^2} E\{W^2\} - \frac{2k}{n} E\{W\} + 1 \right] && \text{(expand the square; linearity)} \\ &= \sigma^4 \left[\frac{k^2(n^2 + 2n)}{n^2} - 2k + 1 \right] = \sigma^4 \left[\frac{k^2(n+2)}{n} - 2k + 1 \right]. && \text{(substitute the } \chi_n^2 \text{ moments)} \end{aligned}$$

This is a parabola in k opening upward; its derivative $2k(n+2)/n - 2$ vanishes at

$$k = \frac{n}{n+2}$$

which is therefore the minimizer, with minimized expected squared error $\sigma^4 \left[1 - \frac{n}{n+2}\right] =$

$$2\sigma^4/(n+2).$$

The point of the question is what k is *not*: it is not 1. Exercise 14's estimator s^2 (the case $k = 1$) is unbiased with expected squared error $2\sigma^4/n$, and shrinking it by the factor $n/(n+2)$ trades a small bias for a variance reduction that wins outright, at every σ^2 and every n (Figure 1). So if expected squared error is the criterion, requiring unbiasedness costs something real: the best unbiased estimator is not the best estimator, and insisting on $E\{d\} = \theta$ amounts to optimizing only the variance half of the bias–variance decomposition. The course has met this trade twice before—the shaded estimator of Assignment 3 and the shrinkage discussion of the notes' §2.10—and this exercise is its cleanest closed-form instance.

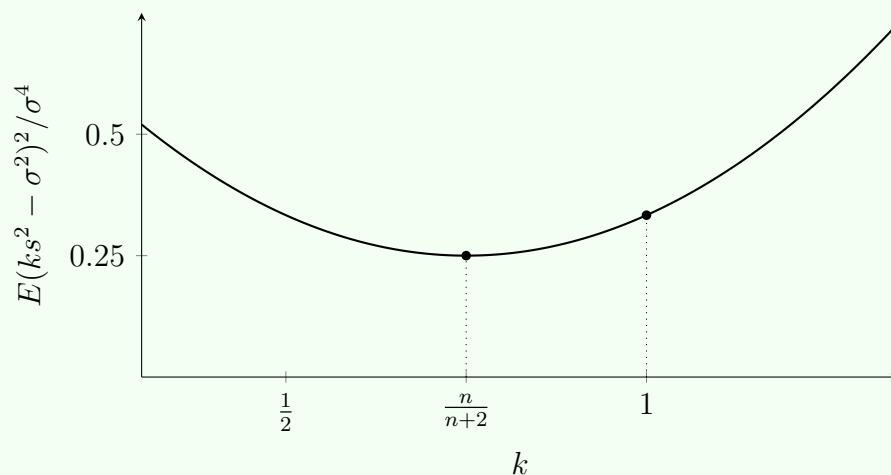


Figure 1. THE COST OF UNBIASEDNESS. Expected squared error of ks^2 as a function of k , drawn for $n = 6$ in units of σ^4 . The unbiased choice $k = 1$ (right dot, error $2/n = 0.333$) sits on the rising branch; the minimizer $k = n/(n+2) = 0.75$ (left dot, error $2/(n+2) = 0.25$) shrinks the estimator toward zero and beats it at every σ^2 .



Exercise 38

(a) How would you estimate $E(1/\sqrt{X})$, based on a random sample of size n , if you did not know the density of X ? (b) Compare your estimate with the maximum likelihood estimate of $E(1/\sqrt{X})$ if X possesses the density $f(x | \theta) = e^{-(x/\theta)}/\theta$, $x > 0$.

SOLUTION:

(a) The quantity is the expectation of the random variable $1/\sqrt{X}$, so estimate it by the corresponding sample average,

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n \frac{1}{\sqrt{X_i}}$$

By linearity, $E\{\hat{g}\} = E\{1/\sqrt{X}\}$ whenever that expectation exists: the estimator is unbiased with no knowledge of the density at all. (This is the principle behind the book's Exercise 37, the sample k th moment estimating the theoretical k th moment; here the function is $x^{-1/2}$ rather than x^k .)

(b) With the exponential density the target has a closed form. Substituting $u = x/\theta$, $dx = \theta du$,

$$\begin{aligned} E\left\{\frac{1}{\sqrt{X}}\right\} &= \int_0^\infty x^{-1/2} \frac{e^{-x/\theta}}{\theta} dx = \frac{1}{\theta} \int_0^\infty (\theta u)^{-1/2} e^{-u} \theta du \\ &= \theta^{-1/2} \int_0^\infty u^{-1/2} e^{-u} du = \frac{\Gamma(\frac{1}{2})}{\sqrt{\theta}} = \sqrt{\frac{\pi}{\theta}}, \end{aligned}$$

using $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. The maximum likelihood estimator of θ in this model is $\bar{X}$ (Assignment 6, Problem 3), and the maximum likelihood estimate of a function of θ is that

function of the estimate, so

$$\widehat{E(1/\sqrt{X})}_{\text{ML}} = \sqrt{\frac{\pi}{\bar{X}}}$$

The comparison the exercise wants: (a) is a general-purpose estimator—unbiased under any parent distribution, but it consults each observation only through $1/\sqrt{X_i}$. (b) commits to the model and channels the whole sample through the one number $\bar{X}$ that the exponential likelihood cares about; it is biased in finite samples (a nonlinear function of an average is not the average of the function), but it converges to the same target and, when the model is right, does so with smaller variance. What is bought with the model assumption is precision; what is risked is everything, since if X is not exponential the second estimator converges to $\sqrt{\pi/EX}$, which is no longer $E(1/\sqrt{X})$. ■

Exercise 54

Find the Bayes estimator of the Poisson parameter μ based on a random sample of size n if $\mathcal{L}(\mu, a) = (a - \mu)^2$ and $\pi(\mu) = e^{-\mu}$, $\mu > 0$.

SOLUTION:

Under squared-error loss the Bayes estimator is the posterior mean (HPS, Thm. 2, p. 38; notes, Thm. 2.9.3). Each observation has pmf $e^{-\mu}\mu^{x_i}/x_i!$, so the likelihood of the sample is

$$L(\mu) = \prod_{i=1}^n \frac{e^{-\mu}\mu^{x_i}}{x_i!} = \frac{e^{-n\mu}\mu^s}{\prod_i x_i!},$$

and the factorials do not involve μ . Matching kernels,

$$\begin{aligned}\pi(\mu \mid x) &\propto e^{-\mu} \cdot e^{-n\mu}\mu^s && \text{(prior times likelihood)} \\ &= \mu^{(1+s)-1} e^{-(n+1)\mu}, && \text{(add exponents)}\end{aligned}$$

the Gamma($1 + s$, $n + 1$) kernel in the shape–rate convention. Its mean is the shape over the rate, so

$$\delta(X) = \frac{1 + \sum_i X_i}{n + 1}$$

This is Assignment 7's Poisson–gamma estimator $(\alpha + s)/(\beta + n)$ at $\alpha = \beta = 1$, the prior $e^{-\mu}$ being exactly Gamma($1, 1$): worth one prior observation carrying one event, and outvoted by the data at rate $1/n$. ■

Exercise 55

Find the Bayes estimator of the parameter θ for the density $f(x | \theta) = \theta e^{-\theta x}$, $x > 0$, based on a single observed value of X , if

$$\mathcal{L}(\theta, a) = (a - \theta)^2 \quad \text{and} \quad \pi(\theta) = 1, \quad 0 < \theta < 1.$$

SOLUTION:

The posterior is proportional to prior times likelihood on the prior's support:

$$\pi(\theta | x) \propto \theta e^{-\theta x}, \quad 0 < \theta < 1,$$

a $\text{Gamma}(2, x)$ kernel cut off at 1—not a standard family, so this time the normalizer and the mean must be computed. Both integrals yield to integration by parts. For the normalizer, with $u = \theta$ and $dv = e^{-\theta x} d\theta$,

$$\int_0^1 \theta e^{-\theta x} d\theta = \left[-\frac{\theta e^{-\theta x}}{x} \right]_0^1 + \frac{1}{x} \int_0^1 e^{-\theta x} d\theta = -\frac{e^{-x}}{x} + \frac{1 - e^{-x}}{x^2} = \frac{1 - (1 + x)e^{-x}}{x^2},$$

and for the numerator, with $u = \theta^2$,

$$\begin{aligned} \int_0^1 \theta^2 e^{-\theta x} d\theta &= \left[-\frac{\theta^2 e^{-\theta x}}{x} \right]_0^1 + \frac{2}{x} \int_0^1 \theta e^{-\theta x} d\theta \\ &= -\frac{e^{-x}}{x} + \frac{2[1 - (1 + x)e^{-x}]}{x^3} = \frac{2 - (x^2 + 2x + 2)e^{-x}}{x^3}, \end{aligned}$$

the last step putting both terms over x^3 : $-x^2 e^{-x} + 2 - 2(1 + x)e^{-x} = 2 - (x^2 + 2x + 2)e^{-x}$.

The Bayes estimator is the ratio,

$$\delta(X) = E\{\theta \mid X\} = \frac{2 - (X^2 + 2X + 2)e^{-X}}{X [1 - (1 + X)e^{-X}]}$$

Two checks confirm the formula behaves. As $x \rightarrow 0$ the posterior tends to $\pi(\theta \mid x) \propto \theta$ on $(0, 1)$, whose mean is $\int_0^1 \theta^2 d\theta / \int_0^1 \theta d\theta = \frac{1/3}{1/2} = \frac{2}{3}$, and the boxed expression has that limit; as $x \rightarrow \infty$ the truncation stops binding and $\delta(x) \approx 2/x$, the mean of the untruncated $\text{Gamma}(2, x)$. A large observation argues for a small rate, and the estimator follows it down—but never past the prior’s floor at 0. ■

Exercise 59(a)

Given $f(x, y) = 8xy$, $0 \leq x \leq 1$, $0 \leq y \leq x$, (a) find the equation of the best predictor $d(x) = E(Y | x)$.

SOLUTION:

The conditional expectation requires the conditional density, which requires the marginal. Integrating the joint over the range of y at fixed x ,

$$f_X(x) = \int_0^x 8xy \, dy = 8x \cdot \frac{x^2}{2} = 4x^3, \quad 0 \leq x \leq 1,$$

so

$$f(y | x) = \frac{f(x, y)}{f_X(x)} = \frac{8xy}{4x^3} = \frac{2y}{x^2}, \quad 0 \leq y \leq x,$$

which integrates to $\int_0^x (2y/x^2) \, dy = x^2/x^2 = 1$, as a conditional density must. Then

$$d(x) = E(Y | x) = \int_0^x y \cdot \frac{2y}{x^2} \, dy = \frac{2}{x^2} \cdot \frac{x^3}{3} = \boxed{\frac{2x}{3}}$$

The best predictor is linear here by accident of the density, not by assumption: the support $0 \leq y \leq x$ scales with x and the conditional density $2y/x^2$ is the same triangular shape on every slice, so its mean sits at the same fraction, two-thirds of the way up. (For this joint density the best linear predictor of the book's Formula (31) therefore coincides with the best predictor—the situation §2.9 of the text describes where $E(Y | x)$ is exactly linear in x .) ■

Exercise 61

Suppose you performed an experiment m times and obtained s successes. Use this information to find the posterior density of p given s if the prior density of p is uniform. If the experiment is performed n additional times and S successes are obtained, what will the Bayes estimate become using squared error if the posterior density of p is treated as the prior density for the second set of experiments? How does this compare with combining the two experiments and using the uniform prior density?

SOLUTION:

First experiment. The uniform prior is $\text{Beta}(1, 1)$, with density 1 on $(0, 1)$, and the likelihood of s successes in m Bernoulli trials is $p^s(1-p)^{m-s}$ (the product $\prod_i p^{x_i}(1-p)^{1-x_i}$ with exponents summed). Matching kernels,

$$\pi(p \mid s) \propto 1 \cdot p^s(1-p)^{m-s} = p^{(1+s)-1}(1-p)^{(1+m-s)-1},$$

so the posterior is

$$p \mid s \sim \text{Beta}(1 + s, 1 + m - s)$$

Second experiment, with this posterior as the new prior. The new likelihood is $p^S(1-p)^{n-S}$, and the exponents add once more:

$$\pi(p \mid s, S) \propto p^{(1+s)-1}(1-p)^{(1+m-s)-1} \cdot p^S(1-p)^{n-S} = p^{(1+s+S)-1}(1-p)^{(1+m-s+n-S)-1},$$

the $\text{Beta}(1 + s + S, 1 + m + n - s - S)$ kernel. Under squared-error loss the Bayes estimate is its mean,

$$\delta = \frac{1 + s + S}{2 + m + n}$$

Combining instead. Pooling the data into one experiment of $m + n$ trials with $s + S$ successes and using the uniform prior gives, by the first computation with (m, s) replaced by $(m + n, s + S)$, the posterior $\text{Beta}(1 + s + S, 1 + (m + n) - (s + S))$ —the same distribution, hence the same estimate. The comparison the exercise asks for:

sequential and batch updating give the same posterior and the same estimate

Nothing about this is special to the Beta family: the sequential posterior is proportional to $\text{prior} \times \text{likelihood}_1 \times \text{likelihood}_2$, and multiplication does not care how the factors are grouped. The posterior after the first experiment is a complete record of what those m trials had to say, which is why yesterday's posterior can serve as today's prior without loss. It is a fitting last exercise for the chapter: the shaded estimator $(1 + s)/(2 + m)$ that entered this course in Assignment 1 as an unexplained formula leaves it as the running total of a Bayesian ledger that can be settled in any order. ■

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